



جامعة أبوظبي
Abu Dhabi University

Department of Computer Science
and Information Technology

Programming Assignment 3

CSC408: Distributed Information Systems

Implementing a Spring Boot-based Secure RESTful API-Based Peer-to-Peer Skills Exchange Platform – SkillSwap[®] UAE

Overview:

In celebration of the UAE's 2025 "Year of Community" initiative under the slogan "**Hand in Hand**," this programming assignment focuses on building an inclusive, collaborative, and skill-sharing digital community. SkillSwap[®]UAE is a platform designed to connect individuals from different backgrounds who are eager to learn new skills and share their expertise with others. By fostering peer-to-peer education, the platform contributes to community growth, mutual support, and lifelong learning.



Students will develop a secure and scalable distributed system that enables users to offer and request skill-based sessions (e.g., language exchange, cooking lessons, coding tutorials, etc.). The platform facilitates session scheduling, feedback collection, and recognition for active contributors.

Assignment Description:

You are to design and implement a secure RESTful API-based Web service using Spring Boot that powers the SkillSwap[®]UAE platform. The platform should allow users to create skill offers, request sessions, and maintain a credit system based on hours exchanged. Additional features such as user matching, feedback ratings, and availability tracking are required for bonus marks.

Requirements:

1. **API Design:** API Design: Craft a RESTful API adhering to REST principles and design best practices. Define clear and intuitive endpoints for user registration, authentication, session submission/completion, and reward tracking.
2. **Endpoints:** Implement endpoints to support the following functionalities:
 - User registration and login (role: learner/mentor/both)
 - Create, update, and browse skill offers
 - Send and respond to session requests
 - Track completed sessions and earned credits
 - Provide and retrieve feedback for completed sessions
3. **Data Model:** Design a suitable data model to represent users, skill offerings, session requests, credit transactions, and feedback records. Use appropriate data structures to efficiently store user profiles, skill details, session scheduling data, and interaction history.

The model should support scalability, data integrity, and clear relationships between users, skills, and exchanged services.

4. **Authentication and Authorization:** Develop secure mechanisms for user authentication to control access to the system. Only authenticated users should be able to create skill listings, request sessions, and participate in credit exchanges.
5. **Reward Calculation:** Define algorithms to calculate rewards based on factors such as the number of skill-sharing sessions completed, the diversity of skills offered, and the positive impact of user feedback and community engagement. Ensure transparency and fairness in the distribution of rewards by providing clear guidelines and visibility into how rewards are calculated.
6. **Security:** Enforce security measures to protect user data and prevent unauthorized access. Implement encryption for sensitive information, sanitize user inputs to prevent injection attacks, and employ secure authentication mechanisms.
7. **Scalability:** Design the system to be scalable, ensuring it can accommodate a growing user base and an increasing volume of skill-sharing sessions and user interactions. Consider implementing distributed storage solutions and caching mechanisms to enhance performance and optimize the handling of data, ensuring quick access and smooth user experience as the platform expands.
8. **Innovation:** The inclusion of advanced functionalities will lead to higher grades. Bonus marks, up to 2.5 marks, may be awarded at the instructor's discretion based on the overall creativity and effectiveness of the advanced solution.
9. **Documentation:** Provide comprehensive documentation for the API, including endpoint descriptions, request/response formats, authentication procedures, and reward calculation algorithms. Include examples and usage guidelines to facilitate integration with client applications.

If you have any questions or need clarification on any aspect of the assignment, feel free to ask. Best of luck, and let's create a collaborative, skill-sharing community that fosters learning and growth together!!



**SKILLSWAP
UAE**
PEER-TO-PEER SKILLS
EXCHANGE PLATFORM

Note: Store all your files in a project called Assig3. In addition, **all file names must include the section, group number, and must indicate Student IDs and names of all members (inside the file)**. For example, if a file is named "Participant.java", then it must be renamed "Participant_2_4" assuming this is for Group 4 of Section 2. **Failure to do that will result in -1 mark penalty.**

Deliverables:

- All projects' files.
- Recorded demonstration showcasing the SkillSwap rewards system and its features.
- A report that includes the following components:
 - a) Documentation detailing the API design, including endpoints, data model, authentication methods, and reward calculation logic.
 - b) Test cases and results demonstrating the functionality and reliability of the API
 - c) Evaluation of the implemented solution, including critiques, and suggestions for potential improvements.

Evaluation:

Your performance in Programming Assignment 3 will be evaluated based **solely on a DEMO and using the below rubric**, during which you must demonstrate how you implemented the requested tasks and showcase the functionality of your program. This assignment is worth **5 marks with EXTRA 2.5 BONUS** towards the final grade of this course. **All your deliverables must be submitted before the deadline of June 1st, 2025, at 23:59, and must be uploaded to Blackboard.** Please note that this deadline is strictly enforced, and late submissions will incur a penalty of -5% per day.

Deadline for the demo: Week of 2-4 June 2025. Students can work in groups of at most three students (NO MORE).

Evaluation Rubric: Please note that this evaluation rubric will be used during the demo, and failure to demonstrate the required functionality and meet the criteria outlined in this rubric will result in a grade of zero for this Programming Assignment. It is essential that you prepare and present a comprehensive demo to showcase your understanding and implementation of the requested tasks.

Description	Maximum
Proper implementation of the RESTful API-Based Web Service. Requirements 1-3	1
Proper implementation of the frontend application using the SkillSwap Rewards service.	0.75
Implementation of the requirements 4-7.	2
Effectiveness of testing and validation efforts.	0.5
Clarity and thoroughness of documentation	0.5
Maintained clear and organized code with comments, proper variable and file naming, and code structure. Clear and well-structured documentation. Effective presentation of results and analysis. Professional formatting and adherence to good formatting practices.	0.25
BONUS: Innovation & Advanced Features	2.5

Total marks: 5 marks + 2.5 as BONUS